

AHMED BOURMECHE

Engineering Student — **Embedded Systems, Industrial IoT & Robotics**

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Open to relocation — Applying for a Final-Year Engineering Internship (PFE)

Professional Profile

Final-year engineering student in Industrial IT and Automation, specialized in **embedded systems, industrial IoT, and robotics**. End-to-end experience across **firmware** and **PCB design, industrial protocols (Modbus, CAN, MQTT), embedded robotics**, and **real-time supervision** of industrial systems.

Technical Skills

Embedded Systems & Hardware: STM32, ESP32 / ESP32-S3, ATmega328P, **embedded C firmware**, KiCad, multilayer PCB design, EMC

Industrial Protocols & Networks: Modbus RTU, CAN, MQTT, RS485/RS232, UART, SPI, I2C

Robotics & AI: Webots, reinforcement learning (PPO, Stable-Baselines3, Gymnasium), SolidWorks, **finite state machines**

Automation & Control: PID, PD control, **MATLAB Simulink, LabVIEW, HIL simulation**, Proteus, **real-time systems**

Backend, Data & DevOps: Python, TypeScript, SQL, Django, FastAPI, PostgreSQL, TimescaleDB, Redis, MySQL, Node.js, Telegraf, React, WebSocket, Docker, Git, CI/CD, Linux

Professional Experience

Industrial IoT Intern — ESD Performance / Vimar SARL

Jun – Sep 2026

Tunisia

- Developed **DCU Edge**, an industrial IoT gateway: multi-device **Modbus RTU** polling (three-phase power meters, air-quality sensors) over **RS-485**, with **MQTT/Telegraf** data ingestion.
- Optimized bus bandwidth via dynamic register pruning, cutting per-device polling cycle time from 4.5 s to **350 ms (13x)** on a shared RS-485 segment.
- Built a UUID-indexed Django/PostgreSQL equipment registry (4 least-privilege roles, CRUD API, and supervision dashboard).
- Engineered a vendor-agnostic Excel-to-JSON integration pipeline (**38 to 241+ registers**); containerized **9 services** with Docker Compose.

Embedded Systems Intern — LAB619 (Industrial IoT Solutions)

Jul – Sep 2025

Tunisia

- Implemented an **industrial IoT node** on **ESP32-S3**: RS485/RS232 and **CAN firmware** for **real-time** data acquisition and remote monitoring.
- Designed an EMC-compliant **4-layer PCB** in **KiCad**, with validated manufacturing files (Gerber, BOM, assembly).

CTO & Co-founder — CyberNexus

2024 – Present

cybernexus.tn

- Sole **DevOps** owner (VPS, **CI/CD**, SSL) for a large-scale, real-time platform (Python, TypeScript, MySQL, **Docker**, Nginx) with **WebSocket** communication; technical lead for a team of 4 engineers.

Lead Developer — Lions Club Tunis INSAT (STEP'ACT 2026)

Jul 2025 – Present

stepact.lionsinsat.tn

- Built a **React/TypeScript/Node.js** registration platform processing **2,500+ registrations**, with real-time participant data management and dynamic multi-distance pricing.

Technical Projects

FARZIT — Autonomous Obstacle-Avoidance Robot

github.com/Bourmeche-Ahmed/FARZIT

- Built a **finite-state-machine** robot on **STM32 Blue Pill**: **ToF** sensing, **PWM/PID motor control**, custom PCB, SolidWorks chassis, with Wi-Fi **ESP32** telemetry streaming (x, y, theta) to a live dashboard.

Autonomous Navigation via Reinforcement Learning

github.com/Bourmeche-Ahmed/RL_NAV_ROBOT

- Trained a **Webots Pioneer 3-AT robot** to navigate via **PPO** (Stable-Baselines3, Gymnasium) using raw **sonar** and proprioceptive data; replaced magnitude-based collision detection with true-position detection: **81% success rate, 0% timeouts** over 50+ randomized episodes.

IIoT Predictive Maintenance Dashboard

github.com/Bourmeche-Ahmed/IIoT-predictive-maintenance-dashboard

- Real-time sensor telemetry platform with **Machine Learning-based failure scoring** and alert thresholds.

PID Control Dashboard — Digital Twin

github.com/Bourmeche-Ahmed/pid-control-dashboard

- MATLAB Simulink digital twin** coupled to a React/Flask dashboard via REST/WebSocket: **real-time PID tuning** and telemetry export, latency **under 100 ms**.

Quarter-Car Suspension System — HIL Simulation

- Modeled and **PD-controlled** the system in **LabVIEW**, coupled to Proteus (ATmega328P) for a **Hardware-in-the-Loop** simulation: **87%** reduction in peak displacement.

Education

Engineering Degree in Industrial IT and Automation — Final Year

2022 – 2027

National Institute of Applied Science and Technology (INSAT), University of Carthage | **Real-Time Systems, Embedded Systems, Industrial Control, Industrial Networks, Robotics**

Languages

Arabic — Native | French — Fluent | English — Fluent